Annual Report - General Order No. R5-2007-0035
Reporting period 01/01/2023 to 12/31/2023.

DAIRY FACILITY INFORMATION

A. NAME OF DAIRY OR BUSINESS OPERATING THE DAIRY: Nissen Dairy, Inc.

Physical address of dairy:

Number and Street	City	County	Zip Code
17263 Brennan AVE	Escalon	San Joaquin	95320

Street and nearest cross street (if no address):

Date facility was originally placed in operation: 01/01/1950

Regional Water Quality Control Board Basin Plan designation: San Joaquin River Basin

County Assessor Parcel Number(s) for dairy facility:

0225-0002-0027-0000	0225-0020-0026-0000	0229-0003-0027-0000	0229-0020-0026-0000	0229-0030-0130-0000	0229-0030-0420-0000

B. OPERATORS

Nissen Dairy, Inc.

Operator name: Nissen Dairy, Inc.

Telephone no.: Landline: ; Cellular: (209) 639-7618

Mailing Address Number and Street	City	State	Zip Code
17283 S Brennan RD	Escalon	CA	95320

This operator is responsible for paying permit fees.

C. OWNERS

Nissen Dairy, Inc.

Legal owner name: Nissen Dairy, Inc.

Telephone no.: Landline: ; Cellular: (209) 639-7618

Mailing Address Number and Street	City	State	Zip Code
17283 S Brennan RD	Escalon	CA	95320

This owner is responsible for paying permit fees.

AVAILABLE NUTRIENTS

A. HERD INFORMATION

	Milk Cows	Dry Cows	Bred Heifers (15-24 mo.)	Heifers (7-14 mo. to breeding)	Calves (4-6 mo.)	Calves (0-3 mo.)
Number open confinement	0	1	200	120	0	0
Number under roof	509	0	0	0	0	0
Maximum number	509	1	200	120	0	0
Average number	509	1	200	120	0	0
Avg live weight (lbs)	1,400	1,450	850	550		

Predominant milk cow breed: Holstein

Average milk production: 68 *pounds per cow per day*

B. MANURE GENERATED

Total manure excreted by the herd: 16,020.22 *tons per reporting period*

Total nitrogen from manure: 199,696.89 *lbs per reporting period*

After ammonia losses (30% loss applied): 139,787.82 *lbs per reporting period*

Total phosphorus from manure: 33,775.98 *lbs per reporting period*

Total potassium from manure: 96,809.37 *lbs per reporting period*

Total salt from manure: 239,892.60 *lbs per reporting period*

C. PROCESS WASTEWATER GENERATED

Process wastewater generated: 375,000 *gallons*

Total nitrogen generated: 4,756.65 *lbs*

Total phosphorus generated: 253.79 *lbs*

Total potassium generated: 9,043.89 *lbs*

Total salt generated: 41,933.63 *lbs*

	375,000	*gallons applied*
+	0	*gallons exported*
-	0	*gallons imported*
=	375,000	*gallons generated*

D. FRESH WATER SOURCES

Source Description	Type
Irrigation District	Surface water

E. SUBSURFACE (TILE) DRAINAGE SOURCES

No subsurface (tile) drainage sources entered.

F. NUTRIENT IMPORTS

No dry manure nutrient imports entered.

No process wastewater nutrient imports entered.

Date	Material type / Description	Quantity	Reporting basis	Moisture (%)	N (%)	P (%)	K (%)	Salt (%)
07/01/2023	Liquid commercial fertilizer *UN32*	14,000.00 *gal*			32.000000	0.000000	0.000000	0.000000

Material type	Total N (lbs)	Total P (lbs)	Total K (lbs)	Total salt (lbs)
Commercial fertilizer / Other	37,385.60	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	0.00	0.00	0.00	0.00
Total imports for all materials	37,385.60	0.00	0.00	0.00

G. NUTRIENT EXPORTS

No solid nutrient exports entered.

No liquid nutrient exports entered.

APPLICATION AREA

A. LIST OF LAND APPLICATION AREAS

Field name	Controlled acres	Cropable acres	Total harvests	Type of waste applied	Parcel number
1	29	29	2	both	X229-X030-XX13-XXXX
2	20	20	2	both	X229-X030-XX42-XXXX
3	25	25	1	none	X225-X020-XX26-XXXX
4	25	25	2	manure	X225-X020-XX26-XXXX
6	18	18	2	manure	X229-X030-XX27-XXXX
7	38	38	2	manure	X225-X020-XX27-XXXX
Totals for areas that were used for application	130	130	10		
Totals for areas that were not used for application	25	25	1		
Land application area totals	155	155	11		

B. CROPS AND HARVESTS

1

Field name: 1

11/02/2022: Oats, silage-soft dough

Crop: Oats, silage-soft dough Acres planted: 29 Plant date: 11/02/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
05/22/2023	440.00 *ton*	Dry-weight		52.9	11,800.00	2,400.00	18,000.00		9.00

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	15.00	150.00	24.00	124.50	0.00
Total actual harvest content	15.17	168.65	34.30	257.26	1,286.32

1

06/20/2023: Corn, silage

Crop: Corn, silage Acres planted: 29 Plant date: 06/20/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
11/01/2023	810.00 *ton*	Dry-weight		62.9	4,200.00	700.00	2,800.00		1.60

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	28.00	224.00	42.00	184.80	0.00
Total actual harvest content	27.93	87.04	14.51	58.03	331.60

2

Field name: 2

11/02/2022: Oats, silage-soft dough

Crop: Oats, silage-soft dough Acres planted: 20 Plant date: 11/02/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
05/22/2023	300.00 *ton*	Dry-weight		67.1	23,000.00	3,500.00	30,600.00		8.90

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	15.00	150.00	24.00	124.50	0.00
Total actual harvest content	15.00	227.01	34.55	302.02	878.43

06/20/2023: Corn, silage

Crop: Corn, silage Acres planted: 20 Plant date: 06/20/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
11/01/2023	560.00 *ton*	Dry-weight		71.2	4,700.00	700.00	5,600.00		2.00

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	28.00	224.00	42.00	184.80	0.00
Total actual harvest content	28.00	75.80	11.29	90.32	322.56

3

Field name: 3

11/01/2010: Pasture

Crop: Pasture Acres planted: 25 Plant date: 11/01/2010

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
12/31/2023	375.00 *ton*	Dry-weight		75.0	20,000.00	5,000.00	20,000.00		5.00

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	15.00	150.00	24.00	124.50	0.00
Total actual harvest content	15.00	150.00	37.50	150.00	375.00

4

Field name: 4

11/02/2022: Oats, silage-soft dough

Crop: Oats, silage-soft dough Acres planted: 25 Plant date: 11/02/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
05/22/2023	380.00 *ton*	Dry-weight		57.0	13,100.00	2,900.00	18,500.00		6.50

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	15.00	150.00	24.00	124.50	0.00
Total actual harvest content	15.20	171.24	37.91	241.83	849.68

06/20/2023: Corn, silage

Crop: Corn, silage Acres planted: 25 Plant date: 06/20/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
11/01/2023	710.00 *ton*	Dry-weight		70.0	3,900.00	500.00	2,700.00		1.60

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	28.00	224.00	42.00	184.80	0.00
Total actual harvest content	28.40	66.46	8.52	46.01	272.64

6

Field name: 6

11/02/2022: Oats, silage-soft dough

Crop: Oats, silage-soft dough Acres planted: 18 Plant date: 11/02/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
05/22/2023	270.00 *ton*	Dry-weight		54.7	9,700.00	2,100.00	16,600.00		7.40

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	15.00	150.00	24.00	124.50	0.00
Total actual harvest content	15.00	131.82	28.54	225.59	1,005.66

06/20/2023: Corn, silage

Crop: Corn, silage Acres planted: 18 Plant date: 06/20/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
11/01/2023	500.00 *ton*	Dry-weight		68.8	4,200.00	800.00	4,800.00		2.00

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	28.00	224.00	42.00	184.80	0.00
Total actual harvest content	27.78	72.80	13.87	83.20	346.67

7

Field name: 7

11/02/2022: Oats, silage-soft dough

Crop: Oats, silage-soft dough Acres planted: 38 Plant date: 11/02/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
05/22/2023	570.00 *ton*	Dry-weight		68.3	10,400.00	2,300.00	16,000.00		6.60

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	15.00	150.00	24.00	124.50	0.00
Total actual harvest content	15.00	98.90	21.87	152.16	627.66

7

06/20/2023: Corn, silage

Crop: Corn, silage Acres planted: 38 Plant date: 06/20/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
11/01/2023	1,070.00 *ton*	Dry-weight		73.9	3,800.00	700.00	5,700.00		1.90

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	28.00	224.00	44.80	184.80	0.00
Total actual harvest content	28.16	55.85	10.29	83.78	279.27

NUTRIENT BUDGET

A. LAND APPLICATIONS

1 - 11/02/2022: Oats, silage-soft dough

Field name: 1

Crop: Oats, silage-soft dough Plant date: 11/02/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/16/2022	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Manure Compost	Compost	82.16	28.79	84.40	0.00	150.00 *ton*
Application event totals		82.16	28.79	84.40	0.00	

1 - 06/20/2023: Corn, silage

Field name: 1

Crop: Corn, silage Plant date: 06/20/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/22/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation District	Surface water	0.00	0.00	0.00	0.00	3,500,883.00 *gal*
Application event totals		0.00	0.00	0.00	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/27/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation District	Surface water	0.00	0.00	0.00	0.00	2,692,987.00 *gal*
Application event totals		0.00	0.00	0.00	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/06/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation District	Surface water	0.00	0.00	0.00	0.00	2,692,987.00 *gal*
Application event totals		0.00	0.00	0.00	0.00	

1 - 06/20/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/16/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation District	Surface water	0.00	0.00	0.00	0.00	2,692,987.00 *gal*
Application event totals		0.00	0.00	0.00	0.00	

08/16/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Wastewater	Process wastewater	98.41	5.25	187.12	867.59	225,000.00 *gal*
Application event totals		98.41	5.25	187.12	867.59	

08/26/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation District	Surface water	0.00	0.00	0.00	0.00	2,692,987.00 *gal*
Application event totals		0.00	0.00	0.00	0.00	

09/05/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation District	Surface water	0.00	0.00	0.00	0.00	2,692,987.00 *gal*
Application event totals		0.00	0.00	0.00	0.00	

09/15/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation District	Surface water	0.00	0.00	0.00	0.00	2,692,987.00 *gal*
Application event totals		0.00	0.00	0.00	0.00	

09/25/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation District	Surface water	0.00	0.00	0.00	0.00	2,692,987.00 *gal*
Application event totals		0.00	0.00	0.00	0.00	

1 - 06/20/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/05/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation District	Surface water	0.00	0.00	0.00	0.00	1,691,987.00 *gal*
Application event totals		0.00	0.00	0.00	0.00	

2 - 11/02/2022: Oats, silage-soft dough

Field name: 2

Crop: Oats, silage-soft dough

Plant date: 11/02/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/16/2022	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Manure Compost	Compost	79.42	27.83	81.59	0.00	100.00 *ton*
Application event totals		79.42	27.83	81.59	0.00	

2 - 06/20/2023: Corn, silage

Field name: 2

Crop: Corn, silage

Plant date: 06/20/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/22/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation District	Surface water	0.00	0.00	0.00	0.00	2,154,384.00 *gal*
Application event totals		0.00	0.00	0.00	0.00	

2 - 06/20/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/27/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation District	Surface water	0.00	0.00	0.00	0.00	1,885,090.00 *gal*
Application event totals		0.00	0.00	0.00	0.00	

07/25/2023	Shank	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation District	Surface water	0.00	0.00	0.00	0.00	1,885,090.00 *gal*
Application event totals		0.00	0.00	0.00	0.00	

08/06/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation District	Surface water	0.00	0.00	0.00	0.00	1,885,090.00 *gal*
Application event totals		0.00	0.00	0.00	0.00	

08/16/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation District	Surface water	0.00	0.00	0.00	0.00	1,885,090.00 *gal*
Application event totals		0.00	0.00	0.00	0.00	

08/16/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Wastewater	Process wastewater	95.13	5.08	180.88	838.67	150,000.00 *gal*
Application event totals		95.13	5.08	180.88	838.67	

08/26/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation District	Surface water	0.00	0.00	0.00	0.00	1,885,090.00 *gal*
Application event totals		0.00	0.00	0.00	0.00	

2 - 06/20/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/05/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation District	Surface water	0.00	0.00	0.00	0.00	1,885,090.00 *gal*
Application event totals		0.00	0.00	0.00	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/15/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation District	Surface water	0.00	0.00	0.00	0.00	1,885,090.00 *gal*
Application event totals		0.00	0.00	0.00	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/25/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation District	Surface water	0.00	0.00	0.00	0.00	1,885,090.00 *gal*
Application event totals		0.00	0.00	0.00	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/05/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation District	Surface water	0.00	0.00	0.00	0.00	1,885,090.00 *gal*
Application event totals		0.00	0.00	0.00	0.00	

3 - 11/01/2010: Pasture

Field name: 3

Crop: Pasture

Plant date: 11/01/2010

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
04/23/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation District	Surface water	0.00	0.00	0.00	0.00	2,154,289.00 *gal*
Application event totals		0.00	0.00	0.00	0.00	

3 - 11/01/2010: Pasture

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/13/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation District	Surface water	0.00	0.00	0.00	0.00	2,154,389.00 *gal*
Application event totals		0.00	0.00	0.00	0.00	

06/02/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation District	Surface water	0.00	0.00	0.00	0.00	2,154,389.00 *gal*
Application event totals		0.00	0.00	0.00	0.00	

06/22/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation District	Surface water	0.00	0.00	0.00	0.00	2,154,389.00 *gal*
Application event totals		0.00	0.00	0.00	0.00	

07/12/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation District	Surface water	0.00	0.00	0.00	0.00	2,154,389.00 *gal*
Application event totals		0.00	0.00	0.00	0.00	

08/01/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation District	Surface water	0.00	0.00	0.00	0.00	2,154,389.00 *gal*
Application event totals		0.00	0.00	0.00	0.00	

08/21/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation District	Surface water	0.00	0.00	0.00	0.00	3,254,389.00 *gal*
Application event totals		0.00	0.00	0.00	0.00	

3 - 11/01/2010: Pasture

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/10/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation District	Surface water	0.00	0.00	0.00	0.00	2,154,389.00 *gal*
Application event totals		0.00	0.00	0.00	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/30/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation District	Surface water	0.00	0.00	0.00	0.00	2,154,389.00 *gal*
Application event totals		0.00	0.00	0.00	0.00	

4 - 11/02/2022: Oats, silage-soft dough

Field name: 4

Crop: Oats, silage-soft dough Plant date: 11/02/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/16/2022	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Manure Compost	Compost	82.60	28.95	84.85	0.00	130.00 *ton*
Application event totals		82.60	28.95	84.85	0.00	

4 - 06/20/2023: Corn, silage

Field name: 4

Crop: Corn, silage Plant date: 06/20/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following

4 - 06/20/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/25/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation District	Surface water	0.00	0.00	0.00	0.00	2,962,285.00 *gal*
Application event totals		0.00	0.00	0.00	0.00	

07/28/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation District	Surface water	0.00	0.00	0.00	0.00	2,423,688.00 *gal*
Application event totals		0.00	0.00	0.00	0.00	

07/26/2023	Shank	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
UN32	Liquid commercial fertilizer	70.00	0.00	0.00	0.00	
Application event totals		70.00	0.00	0.00	0.00	

08/07/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation District	Surface water	0.00	0.00	0.00	0.00	2,423,688.00 *gal*
Application event totals		0.00	0.00	0.00	0.00	

08/17/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation District	Surface water	0.00	0.00	0.00	0.00	2,423,688.00 *gal*
Application event totals		0.00	0.00	0.00	0.00	

08/27/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation District	Surface water	0.00	0.00	0.00	0.00	2,423,688.00 *gal*
Application event totals		0.00	0.00	0.00	0.00	

4 - 06/20/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/06/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation District	Surface water	0.00	0.00	0.00	0.00	2,423,688.00 *gal*
Application event totals		0.00	0.00	0.00	0.00	

09/16/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation District	Surface water	0.00	0.00	0.00	0.00	2,423,688.00 *gal*
Application event totals		0.00	0.00	0.00	0.00	

09/26/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation District	Surface water	0.00	0.00	0.00	0.00	2,423,688.00 *gal*
Application event totals		0.00	0.00	0.00	0.00	

10/06/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation District	Surface water	0.00	0.00	0.00	0.00	2,423,688.00 *gal*
Application event totals		0.00	0.00	0.00	0.00	

6 - 11/02/2022: Oats, silage-soft dough

Field name: 6

Crop: Oats, silage-soft dough Plant date: 11/02/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/16/2022	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Manure Compost	Compost	79.42	27.83	81.59	0.00	90.00 *ton*
Application event totals		79.42	27.83	81.59	0.00	

6 - 06/20/2023: Corn, silage

Field name: 6

Crop: Corn, silage

Plant date: 06/20/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/23/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation District	Surface water	0.00	0.00	0.00	0.00	2,154,389.00 *gal*
Application event totals		0.00	0.00	0.00	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/28/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation District	Surface water	0.00	0.00	0.00	0.00	1,885,090.00 *gal*
Application event totals		0.00	0.00	0.00	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/26/2023	Shank	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
UN32	Liquid commercial fertilizer	70.00	0.00	0.00	0.00	
Application event totals		70.00	0.00	0.00	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/07/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation District	Surface water	0.00	0.00	0.00	0.00	1,615,792.00 *gal*
Application event totals		0.00	0.00	0.00	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/17/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation District	Surface water	0.00	0.00	0.00	0.00	1,615,792.00 *gal*
Application event totals		0.00	0.00	0.00	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/27/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation District	Surface water	0.00	0.00	0.00	0.00	1,615,792.00 *gal*
Application event totals		0.00	0.00	0.00	0.00	

6 - 06/20/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/06/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation District	Surface water	0.00	0.00	0.00	0.00	2,625,792.00 *gal*
Application event totals		0.00	0.00	0.00	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/16/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation District	Surface water	0.00	0.00	0.00	0.00	1,615,792.00 *gal*
Application event totals		0.00	0.00	0.00	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/26/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation District	Surface water	0.00	0.00	0.00	0.00	1,615,792.00 *gal*
Application event totals		0.00	0.00	0.00	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/06/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation District	Surface water	0.00	0.00	0.00	0.00	1,615,792.00 *gal*
Application event totals		0.00	0.00	0.00	0.00	

7 - 11/02/2022: Oats, silage-soft dough

Field name: 7

Crop: Oats, silage-soft dough

Plant date: 11/02/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/16/2022	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Manure Compost	Compost	79.42	27.83	81.59	0.00	190.00 *ton*
Application event totals		79.42	27.83	81.59	0.00	

7 - 06/20/2023: Corn, silage

Field name: 7

Crop: Corn, silage

Plant date: 06/20/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/21/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation District	Surface water	0.00	0.00	0.00	0.00	4,039,480.00 *gal*
Application event totals		0.00	0.00	0.00	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/25/2023	Shank	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
UN32	Liquid commercial fertilizer	70.00	0.00	0.00	0.00	
Application event totals		70.00	0.00	0.00	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/26/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation District	Surface water	0.00	0.00	0.00	0.00	3,500,883.00 *gal*
Application event totals		0.00	0.00	0.00	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/05/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation District	Surface water	0.00	0.00	0.00	0.00	3,500,883.00 *gal*
Application event totals		0.00	0.00	0.00	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/15/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation District	Surface water	0.00	0.00	0.00	0.00	3,500,883.00 *gal*
Application event totals		0.00	0.00	0.00	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/25/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation District	Surface water	0.00	0.00	0.00	0.00	3,500,883.00 *gal*
Application event totals		0.00	0.00	0.00	0.00	

7 - 06/20/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/04/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation District	Surface water	0.00	0.00	0.00	0.00	3,500,883.00 *gal*
Application event totals		0.00	0.00	0.00	0.00	

09/14/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation District	Surface water	0.00	0.00	0.00	0.00	3,500,883.00 *gal*
Application event totals		0.00	0.00	0.00	0.00	

09/24/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation District	Surface water	0.00	0.00	0.00	0.00	3,500,883.00 *gal*
Application event totals		0.00	0.00	0.00	0.00	

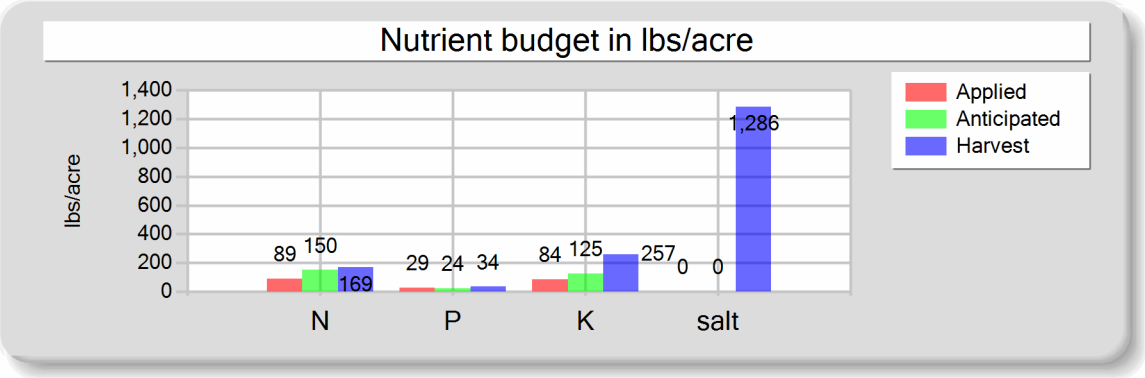
10/04/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation District	Surface water	0.00	0.00	0.00	0.00	3,500,883.00 *gal*
Application event totals		0.00	0.00	0.00	0.00	

B. NUTRIENT BUDGET

1 - 11/02/2022: Oats, silage-soft dough

Field name: 1 Crop: Oats, silage-soft dough Plant date: 11/02/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	82.16	28.79	84.40	0.00
Process wastewater	0.00	0.00	0.00	0.00
Fresh water	0.00	0.00	0.00	0.00
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	89.16	28.79	84.40	0.00
Anticipated crop nutrient removal	150.00	24.00	124.50	0.00
Actual crop nutrient removal	168.65	34.30	257.26	1,286.32
Nutrient balance	-79.49	-5.51	-172.86	-1,286.32
Applied to removed ratio	0.53	0.84	0.33	0.00

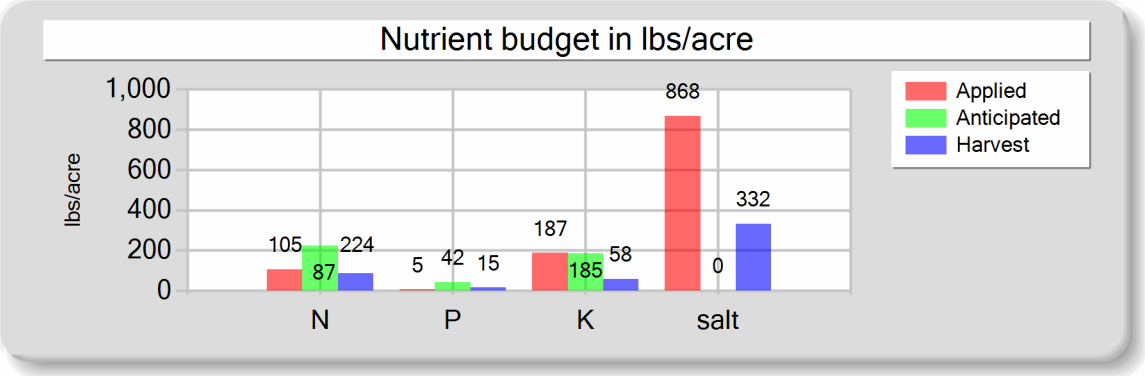
Fresh water applied
0.00 *gallons*
0.00 *acre-inches*
0.00 *inches/acre*

Process wastewater applied
0.00 *gallons*
0.00 *acre-inches*
0.00 *inches/acre*

Total harvests for the crop
1 *harvests*

1 - 06/20/2023: Corn, silage

Field name: 1 Crop: Corn, silage Plant date: 06/20/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	98.41	5.25	187.12	867.59
Fresh water	0.00	0.00	0.00	0.00
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	105.41	5.25	187.12	867.59
Anticipated crop nutrient removal	224.00	42.00	184.80	0.00
Actual crop nutrient removal	87.04	14.51	58.03	331.60
Nutrient balance	18.37	-9.26	129.09	536.00
Applied to removed ratio	1.21	0.36	3.22	2.62

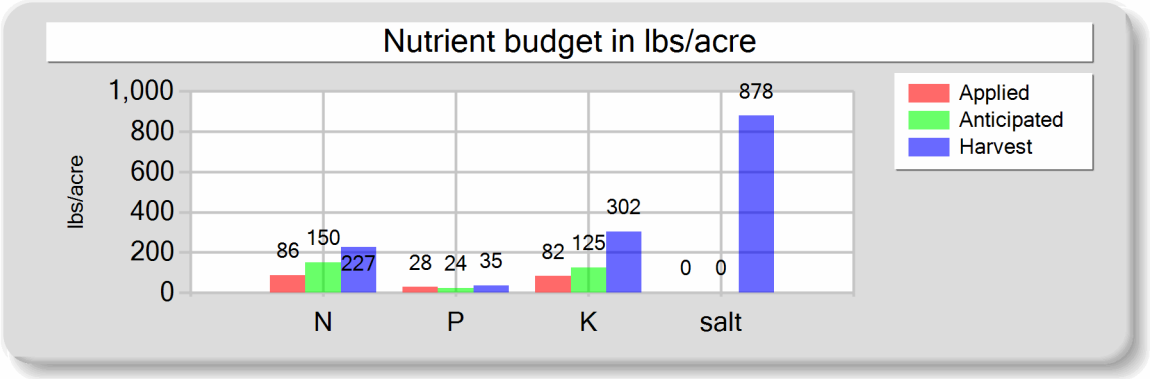
Fresh water applied	
24,043,779.00	*gallons*
885.45	*acre-inches*
30.53	*inches/acre*

Process wastewater applied	
225,000.00	*gallons*
8.29	*acre-inches*
0.29	*inches/acre*

Total harvests for the crop	
1	*harvests*

2 - 11/02/2022: Oats, silage-soft dough

Field name: 2 Crop: Oats, silage-soft dough Plant date: 11/02/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	79.42	27.83	81.59	0.00
Process wastewater	0.00	0.00	0.00	0.00
Fresh water	0.00	0.00	0.00	0.00
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	86.42	27.83	81.59	0.00
Anticipated crop nutrient removal	150.00	24.00	124.50	0.00
Actual crop nutrient removal	227.01	34.55	302.02	878.43
Nutrient balance	-140.59	-6.71	-220.44	-878.43
Applied to removed ratio	0.38	0.81	0.27	0.00

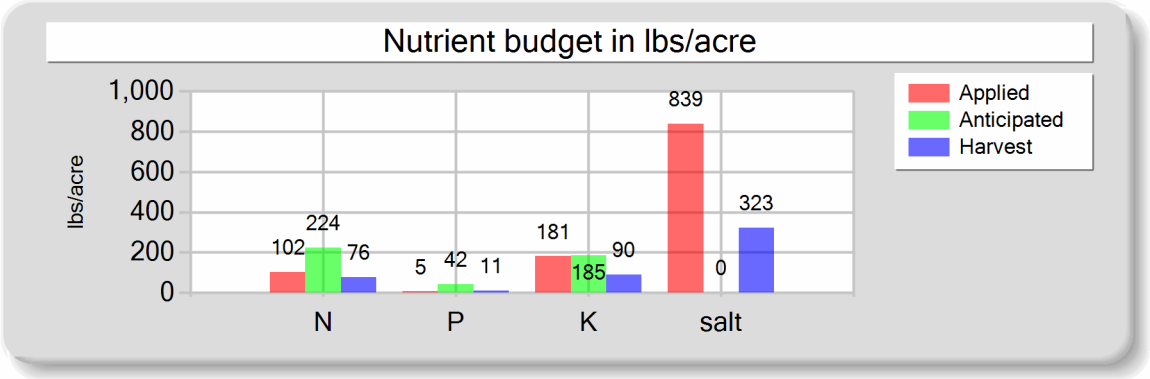
Fresh water applied
0.00 *gallons*
0.00 *acre-inches*
0.00 *inches/acre*

Process wastewater applied
0.00 *gallons*
0.00 *acre-inches*
0.00 *inches/acre*

Total harvests for the crop
1 *harvests*

2 - 06/20/2023: Corn, silage

Field name: 2 Crop: Corn, silage Plant date: 06/20/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	95.13	5.08	180.88	838.67
Fresh water	0.00	0.00	0.00	0.00
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	102.13	5.08	180.88	838.67
Anticipated crop nutrient removal	224.00	42.00	184.80	0.00
Actual crop nutrient removal	75.80	11.29	90.32	322.56
Nutrient balance	26.33	-6.21	90.56	516.11
Applied to removed ratio	1.35	0.45	2.00	2.60

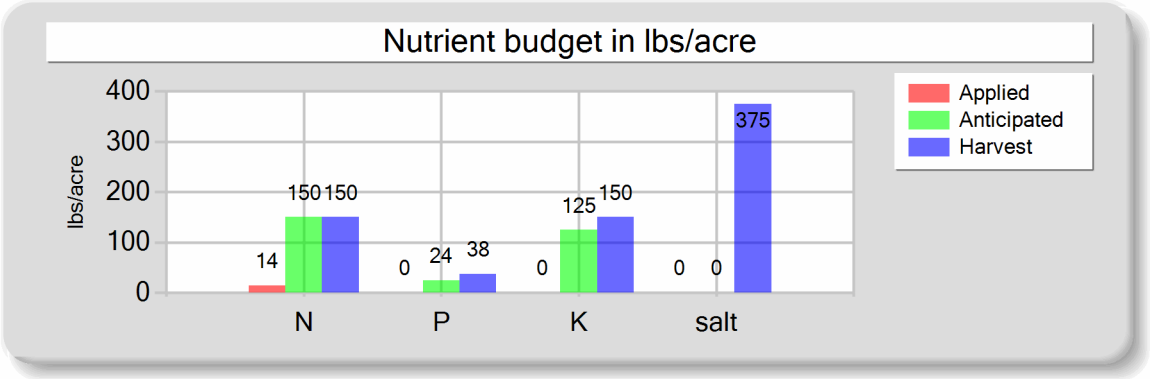
Fresh water applied	
19,120,194.00	*gallons*
704.13	*acre-inches*
35.21	*inches/acre*

Process wastewater applied	
150,000.00	*gallons*
5.52	*acre-inches*
0.28	*inches/acre*

Total harvests for the crop	
1	*harvests*

3 - 11/01/2010: Pasture

Field name: 3 Crop: Pasture Plant date: 11/01/2010



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	0.00	0.00	0.00	0.00
Fresh water	0.00	0.00	0.00	0.00
Atmospheric deposition	14.00	0.00	0.00	0.00
Total nutrients applied	14.00	0.00	0.00	0.00
Anticipated crop nutrient removal	150.00	24.00	124.50	0.00
Actual crop nutrient removal	150.00	37.50	150.00	375.00
Nutrient balance	-136.00	-37.50	-150.00	-375.00
Applied to removed ratio	0.09	0.00	0.00	0.00

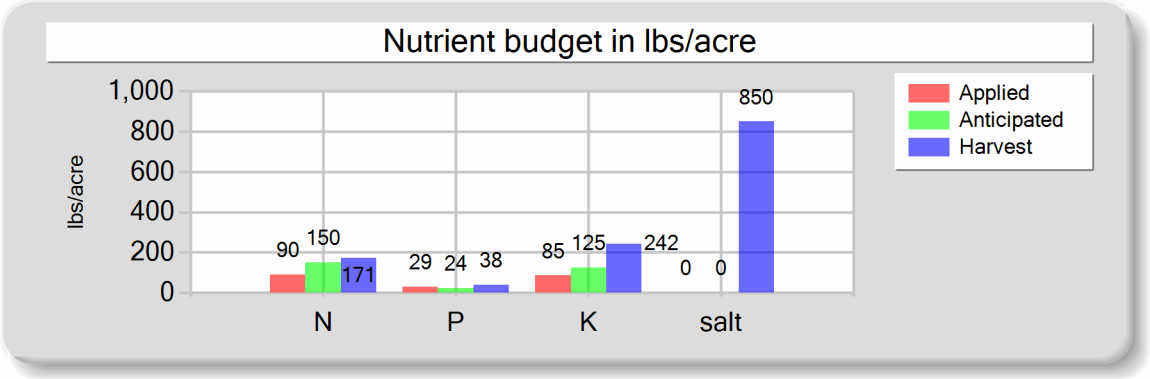
Fresh water applied	
20,489,401.00	*gallons*
754.55	*acre-inches*
30.18	*inches/acre*

Process wastewater applied	
0.00	*gallons*
0.00	*acre-inches*
0.00	*inches/acre*

Total harvests for the crop	
1	*harvests*

4 - 11/02/2022: Oats, silage-soft dough

Field name: 4 Crop: Oats, silage-soft dough Plant date: 11/02/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	82.60	28.95	84.85	0.00
Process wastewater	0.00	0.00	0.00	0.00
Fresh water	0.00	0.00	0.00	0.00
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	89.60	28.95	84.85	0.00
Anticipated crop nutrient removal	150.00	24.00	124.50	0.00
Actual crop nutrient removal	171.24	37.91	241.83	849.68
Nutrient balance	-81.65	-8.96	-156.98	-849.68
Applied to removed ratio	0.52	0.76	0.35	0.00

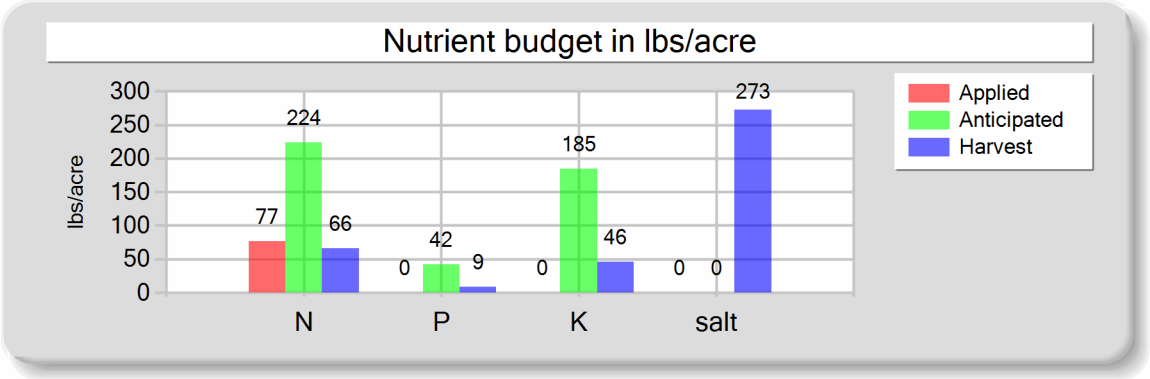
Fresh water applied
0.00 *gallons*
0.00 *acre-inches*
0.00 *inches/acre*

Process wastewater applied
0.00 *gallons*
0.00 *acre-inches*
0.00 *inches/acre*

Total harvests for the crop
1 *harvests*

4 - 06/20/2023: Corn, silage

Field name: 4 Crop: Corn, silage Plant date: 06/20/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	70.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	0.00	0.00	0.00	0.00
Fresh water	0.00	0.00	0.00	0.00
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	77.00	0.00	0.00	0.00
Anticipated crop nutrient removal	224.00	42.00	184.80	0.00
Actual crop nutrient removal	66.46	8.52	46.01	272.64
Nutrient balance	10.54	-8.52	-46.01	-272.64
Applied to removed ratio	1.16	0.00	0.00	0.00

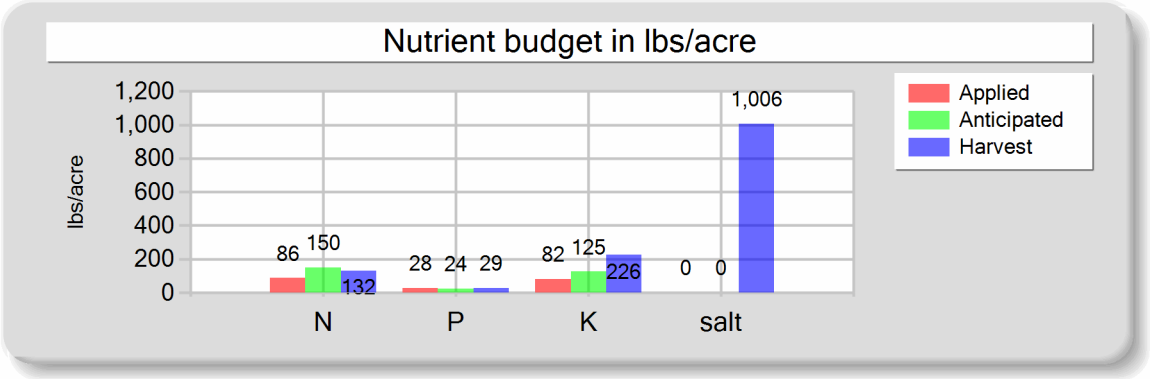
Fresh water applied	
22,351,789.00	*gallons*
823.14	*acre-inches*
32.93	*inches/acre*

Process wastewater applied	
0.00	*gallons*
0.00	*acre-inches*
0.00	*inches/acre*

Total harvests for the crop	
1	*harvests*

6 - 11/02/2022: Oats, silage-soft dough

Field name: 6 Crop: Oats, silage-soft dough Plant date: 11/02/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	79.42	27.83	81.59	0.00
Process wastewater	0.00	0.00	0.00	0.00
Fresh water	0.00	0.00	0.00	0.00
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	86.42	27.83	81.59	0.00
Anticipated crop nutrient removal	150.00	24.00	124.50	0.00
Actual crop nutrient removal	131.82	28.54	225.59	1,005.66
Nutrient balance	-45.40	-0.71	-144.01	-1,005.66
Applied to removed ratio	0.66	0.98	0.36	0.00

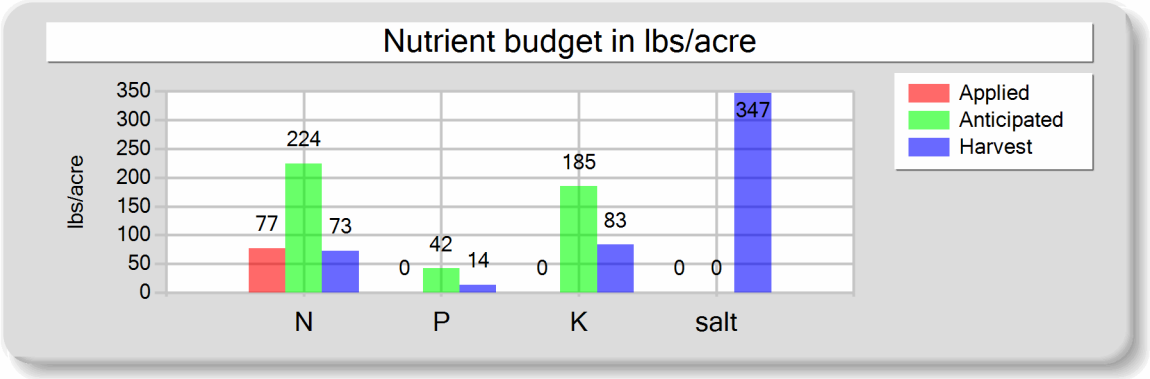
Fresh water applied
0.00 *gallons*
0.00 *acre-inches*
0.00 *inches/acre*

Process wastewater applied
0.00 *gallons*
0.00 *acre-inches*
0.00 *inches/acre*

Total harvests for the crop
1 *harvests*

6 - 06/20/2023: Corn, silage

Field name: 6 Crop: Corn, silage Plant date: 06/20/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	70.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	0.00	0.00	0.00	0.00
Fresh water	0.00	0.00	0.00	0.00
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	77.00	0.00	0.00	0.00
Anticipated crop nutrient removal	224.00	42.00	184.80	0.00
Actual crop nutrient removal	72.80	13.87	83.20	346.67
Nutrient balance	4.20	-13.87	-83.20	-346.67
Applied to removed ratio	1.06	0.00	0.00	0.00

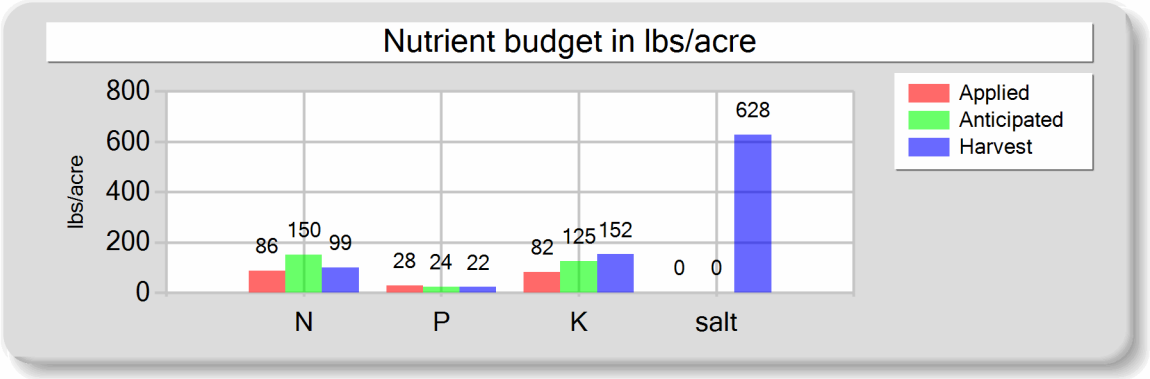
Fresh water applied	
16,360,023.00	*gallons*
602.48	*acre-inches*
33.47	*inches/acre*

Process wastewater applied	
0.00	*gallons*
0.00	*acre-inches*
0.00	*inches/acre*

Total harvests for the crop	
1	*harvests*

7 - 11/02/2022: Oats, silage-soft dough

Field name: 7 Crop: Oats, silage-soft dough Plant date: 11/02/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	79.42	27.83	81.59	0.00
Process wastewater	0.00	0.00	0.00	0.00
Fresh water	0.00	0.00	0.00	0.00
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	86.42	27.83	81.59	0.00
Anticipated crop nutrient removal	150.00	24.00	124.50	0.00
Actual crop nutrient removal	98.90	21.87	152.16	627.66
Nutrient balance	-12.48	5.96	-70.57	-627.66
Applied to removed ratio	0.87	1.27	0.54	0.00

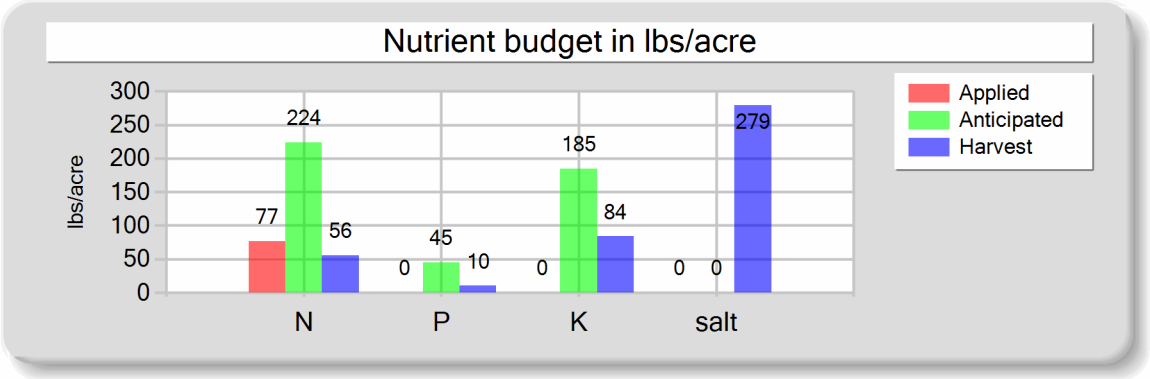
Fresh water applied
0.00 *gallons*
0.00 *acre-inches*
0.00 *inches/acre*

Process wastewater applied
0.00 *gallons*
0.00 *acre-inches*
0.00 *inches/acre*

Total harvests for the crop
1 *harvests*

7 - 06/20/2023: Corn, silage

Field name: 7 Crop: Corn, silage Plant date: 06/20/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	70.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	0.00	0.00	0.00	0.00
Fresh water	0.00	0.00	0.00	0.00
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	77.00	0.00	0.00	0.00
Anticipated crop nutrient removal	224.00	44.80	184.80	0.00
Actual crop nutrient removal	55.85	10.29	83.78	279.27
Nutrient balance	21.15	-10.29	-83.78	-279.27
Applied to removed ratio	1.38	0.00	0.00	0.00

Fresh water applied	
32,046,544.00	*gallons*
1,180.17	*acre-inches*
31.06	*inches/acre*

Process wastewater applied	
0.00	*gallons*
0.00	*acre-inches*
0.00	*inches/acre*

Total harvests for the crop	
1	*harvests*

NUTRIENT ANALYSES

A. MANURE ANALYSES

Denele Analytical, Inc.

Sample and source description: Denele Analytical, Inc.
Sample date: 08/04/2022 Material type: Compost Source of analysis: Lab analysis Method of reporting: Dry-weight
Moisture: 63.9 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Calcium (mg/kg)	Magnesium (mg/kg)	Sodium (mg/kg)	Sulfur (mg/kg)	Chloride (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	22,000.00	7,710.00	22,600.00							
DL	100.00	10.00	100.00							

Denele Analytical, Inc.

Sample and source description: Denele Analytical, Inc.
Sample date: 01/26/2023 Material type: Corral solids Source of analysis: Lab analysis Method of reporting: Dry-weight
Moisture: 28.5 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Calcium (mg/kg)	Magnesium (mg/kg)	Sodium (mg/kg)	Sulfur (mg/kg)	Chloride (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	23,200.00	10,300.00	31,800.00							
DL	100.00	100.00	100.00							

Denele Analytical, Inc.

Sample and source description: Denele Analytical, Inc.
Sample date: 08/01/2023 Material type: Compost Source of analysis: Lab analysis Method of reporting: Dry-weight
Moisture: 7.6 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Calcium (mg/kg)	Magnesium (mg/kg)	Sodium (mg/kg)	Sulfur (mg/kg)	Chloride (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	18,400.00	5,550.00	22,800.00							
DL	100.00	10.00	100.00							

B. PROCESS WASTEWATER ANALYSES

Denele Analytical, Inc.

Sample and source description: Denele Analytical, Inc.

Sample date: 01/26/2023 Material type: Process wastewater Source of analysis: Lab analysis pH:

	Kjeldahl-N (mg/L)	NH4-N (mg/L)	NH3-N (mg/L)	Nitrate-N (mg/L)	Total P (mg/L)	Total K (mg/L)	Calcium (mg/L)	Magnes. (mg/L)	Sodium (mg/L)	Bicarb. (mg/L)	Carb. (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	2,690.00	935.00	0.00	0.00	216.00	1,910.00								12,000.00	5,560
DL	1.00	1.00	1.00	1.00	1.00	1.00								100.00	10

Denele Analytical, Inc.

Sample and source description: Denele Analytical, Inc.

Sample date: 08/01/2023 Material type: Process wastewater Source of analysis: Lab analysis pH:

	Kjeldahl-N (mg/L)	NH4-N (mg/L)	NH3-N (mg/L)	Nitrate-N (mg/L)	Total P (mg/L)	Total K (mg/L)	Calcium (mg/L)	Magnes. (mg/L)	Sodium (mg/L)	Bicarb. (mg/L)	Carb. (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	1,520.00	176.00	0.00	0.00	81.10	2,890.00								20,000.00	13,400
DL	1.00	1.00	1.00	1.00	0.10	1.00								100.00	100

C. FRESH WATER ANALYSES

Irrigation District

IEH-JL Analytical

Sample description: IEH-JL Analytical

Sample date: 06/14/2023 Source of analysis: Lab analysis

	Total N (mg/L)	NH4-N (mg/L)	Nitrate-N (mg/L)	Calcium (mg/L)	Magnesium (mg/L)	Sodium (mg/L)	Bicarbonate (mg/L)	Carbonate (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	0.00	0.00	0.00								0.00	0
DL	2.00	2.00	0.40								1.00	20

D. SOIL ANALYSES

No soil analyses entered.

E. PLANT TISSUE ANALYSES

1 - 11/02/2022: Oats, silage-soft dough

IEH-JL Analytical

Sample and source description: IEH-JL Analytical

Sample date: 05/22/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 52.9 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	11,800.00	2,400.00	18,000.00		9.00
DL	100.00	100.00	100.00		0.10

1 - 06/20/2023: Corn, silage

IEH-JL Analytical

Sample and source description: IEH-JL Analytical

Sample date: 11/01/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 62.9 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	4,200.00	700.00	2,800.00		1.60
DL	100.00	100.00	100.00		0.10

2 - 11/02/2022: Oats, silage-soft dough

IEH-JL Analytical

Sample and source description: IEH-JL Analytical

Sample date: 05/22/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 67.1 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	23,000.00	3,500.00	30,600.00		8.90
DL	100.00	100.00	100.00		0.10

2 - 06/20/2023: Corn, silage

IEH-JL Analytical

Sample and source description: IEH-JL Analytical

Sample date: 11/01/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 71.2 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	4,700.00	700.00	5,600.00		2.00
DL	100.00	100.00	100.00		0.10

3 - 11/01/2010: Pasture

Estimate - Not Sampled

Sample and source description: Estimate - Not Sampled

Sample date: 12/31/2023 Source of analysis: Other/estimated Method of reporting: Dry-weight

Moisture: 75.0 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	20,000.00	5,000.00	20,000.00		5.00
DL	100.00	100.00	100.00		0.10

4 - 11/02/2022: Oats, silage-soft dough

IEH-JL Analytical

Sample and source description: IEH-JL Analytical

Sample date: 05/22/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 57.0 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	13,100.00	2,900.00	18,500.00		6.50
DL	100.00	100.00	100.00		0.10

4 - 06/20/2023: Corn, silage

IEH-JL Analytical

Sample and source description: IEH-JL Analytical

Sample date: 11/01/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 70.0 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	3,900.00	500.00	2,700.00		1.60
DL	100.00	100.00	100.00		0.10

6 - 11/02/2022: Oats, silage-soft dough

IEH-JL Analytical

Sample and source description: IEH-JL Analytical

Sample date: 05/22/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 54.7 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	9,700.00	2,100.00	16,600.00		7.40
DL	100.00	100.00	100.00		0.10

6 - 06/20/2023: Corn, silage

IEH-JL Analytical

Sample and source description: IEH-JL Analytical

Sample date: 11/01/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 68.8 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	4,200.00	800.00	4,800.00		2.00
DL	100.00	100.00	100.00		0.10

7 - 11/02/2022: Oats, silage-soft dough

IEH-JL Analytical

Sample and source description: IEH-JL Analytical

Sample date: 05/22/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 68.3 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	10,400.00	2,300.00	16,000.00		6.60
DL	100.00	100.00	100.00		0.10

7 - 06/20/2023: Corn, silage

IEH-JL Analytical

Sample and source description: IEH-JL Analytical

Sample date: 11/01/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 73.9 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	3,800.00	700.00	5,700.00		1.90
DL	100.00	100.00	100.00		0.10

F. SUBSURFACE (TILE) DRAINAGE ANALYSES

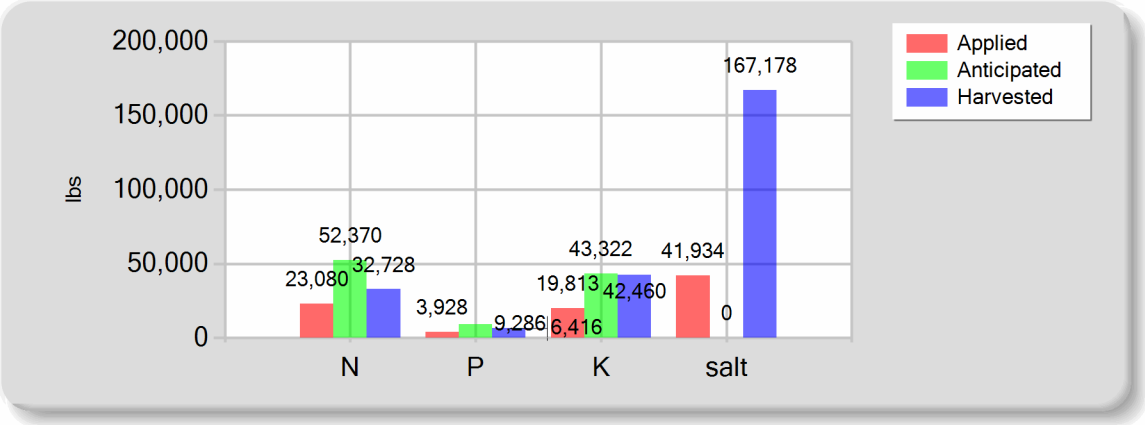
No subsurface (tile) drainage analyses entered.

NUTRIENT APPLICATIONS, POTENTIAL REMOVAL, AND BALANCE

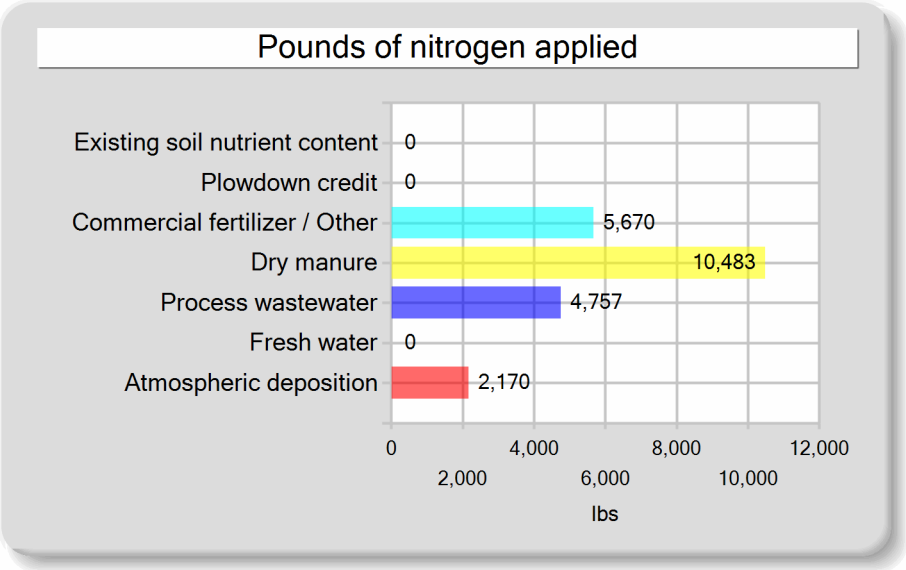
A. SUMMARY OF NUTRIENT APPLICATIONS, POTENTIAL REMOVAL, AND BALANCE

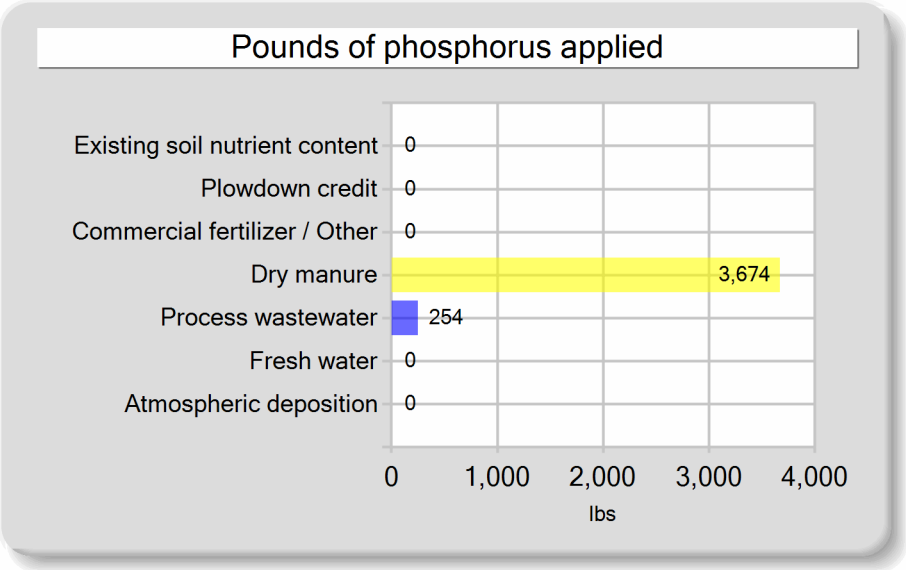
	Total N (lbs)	Total P (lbs)	Total K (lbs)	Total salt (lbs)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	5,670.00	0.00	0.00	0.00
Dry manure	10,483.44	3,673.97	10,769.35	0.00
Process wastewater	4,756.65	253.79	9,043.89	41,933.63
Fresh water	0.00	0.00	0.00	0.00
Atmospheric deposition	2,170.00	0.00	0.00	0.00
Total nutrients applied	23,080.09	3,927.76	19,813.25	41,933.63
Anticipated crop nutrient removal	52,370.00	9,286.40	43,321.50	0.00
Actual crop nutrient removal	32,727.88	6,415.83	42,460.32	167,177.54
Nutrient balance	-9,647.79	-2,488.07	-22,647.08	-125,243.92
Applied to removed ratio	0.71	0.61	0.47	0.25

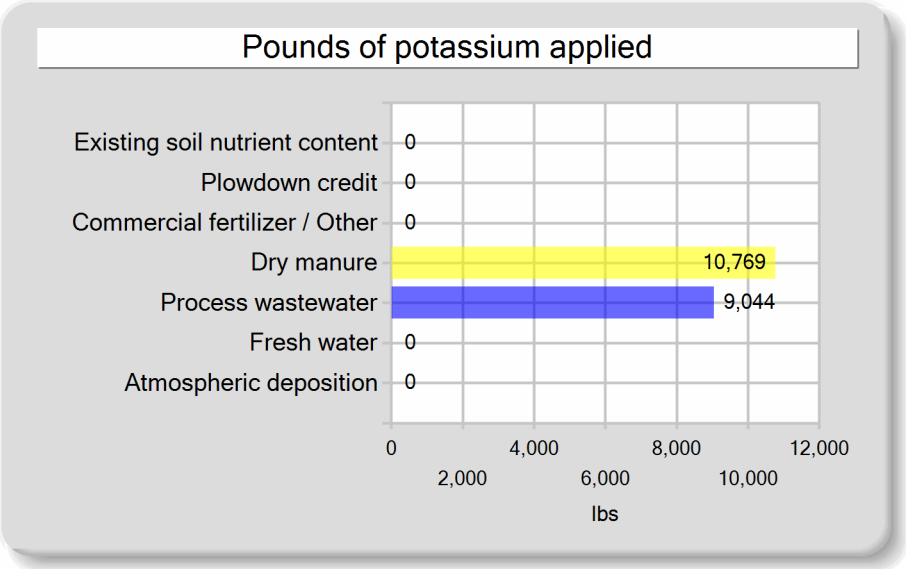
B. POUNDS OF NUTRIENT APPLIED VS. CROP REMOVAL

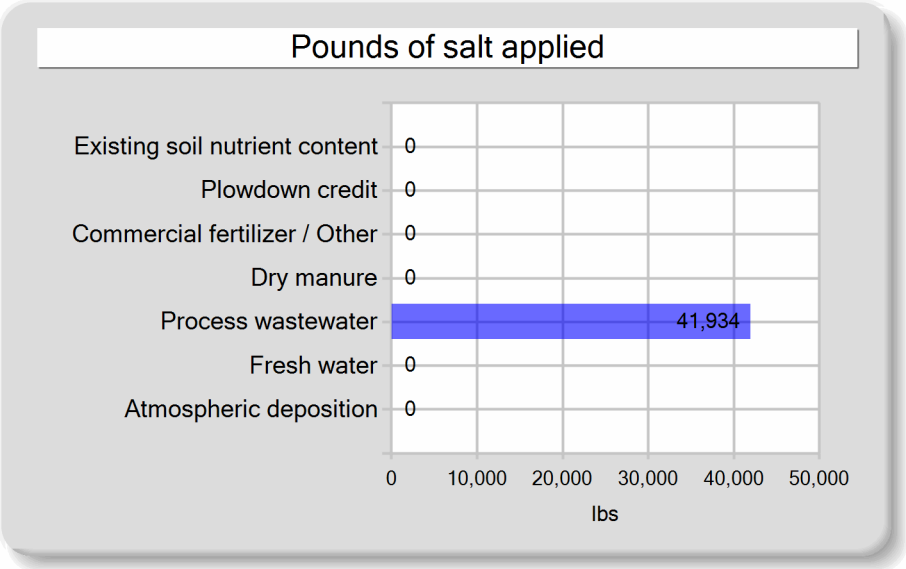


C. POUNDS OF NUTRIENT APPLIED BY MATERIAL TYPE









EXCEPTION REPORTING

A. MANURE, PROCESS WASTEWATER, AND OTHER DAIRY WASTE DISCHARGES

The following is a summary of all manure and process wastewater discharges from the production area to surface water or to land areas (land application areas or otherwise) when not in accordance with the facility's Nutrient Management Plan.

No manure or process wastewater discharges occurred during the reporting period.

B. STORM WATER DISCHARGES

The following is a summary of all storm water discharges from the production area to surface water during the reporting period when not in accordance with the facility 's Nutrient Management Plan.

No stormwater discharges occurred during the reporting period.

C. LAND APPLICATION AREA TO SURFACE WATER DISCHARGES

The following is a summary of all discharges from the land application area to surface water that have occurred during the reporting period when not in accordance with the facility's Nutrient Management Plan.

No land application area to surface water discharges occurred during the reporting period.

NUTRIENT MANAGEMENT PLAN AND EXPORT AGREEMENT STATEMENTS

A. NUTRIENT MANAGEMENT PLAN STATEMENTS

Was the facility's NMP updated in the reporting period?	No
Was the facility's NMP developed by a certified nutrient management planner (specialist) as specified in Attachment C of the General Order?	Yes
Was the facility's NMP approved by a certified nutrient management planner (specialist) as specified in Attachment C of the General Order?	Yes

B. EXPORT AGREEMENT STATEMENT

Are there any written agreements with third parties to receive manure or process wastewater that are new or were revised within the reporting period?	No

ADDITIONAL NOTES

A. NOTES

No notes entered for this annual report.

CERTIFICATION

A. OWNER AND/OR OPERATOR CERTIFICATION

I certify under penalty of law that I have personally examined and am familiar with the information submitted in this document and all attachments and that, based on my inquiry of those individuals immediately responsible for obtaining the information, I believe that the information is true, accurate, and complete. I am aware that there are significant penalties for submitting false information, including the possibility of fine and imprisonment.

SIGNATURE OF OWNER OF FACILITY	SIGNATURE OF OPERATOR OF FACILITY
Nissen Dairy, Inc.	SAME AS OWNER
PRINT OR TYPE NAME	PRINT OR TYPE NAME
DATE	DATE

ATTACHMENTS

A. REQUIRED ATTACHMENTS

The following lists the required documents that should be attached to the Annual Report when submitted .

Annual Dairy Facility Assessment
Provide an Annual Dairy Facility Assessment (an update to the Preliminary Dairy Facility Assessment in Attachment A) for each reporting period. On the PDFA Final page, click on the ADFA Report button to generate an ADFA report after updating information as needed .

Manure/Process Wastewater Tracking Manifests
Provide copies of all manure/process wastewater tracking manifests for the reporting period, signed by both the owner/operator and the hauler.

Corrective Actions Documents
Provide records documenting any corrective actions taken to correct deficiencies noted as a result of the inspections required in the Monitoring Requirements of the General Order. Deficiencies not corrected in 30 days must be accompanied by an explanation of the factors preventing immediate correction.

Groundwater Monitoring
Dischargers that monitor supply wells or subsurface (tile) drainage systems, or that have monitoring well systems must submit monitoring results as directed in the General Order, Groundwater Reporting Section starting on page MRP-13.

Storm Water Monitoring
Dischargers that are required to monitor storm water more frequently than required in the General Order must submit monitoring results as directed in the General Order, Storm Water Reporting Section on page MRP-14.

Annual Report - General Order No. R5-2007-0035
Reporting period 01/01/2023 to 12/31/2023.

CERTIFICATION

A. OWNER AND/OR OPERATOR CERTIFICATION

I certify under penalty of law that I have personally examined and am familiar with the information submitted in this document and all attachments and that, based on my inquiry of those individuals immediately responsible for obtaining the information, I believe that the information is true, accurate, and complete. I am aware that there are significant penalties for submitting false information, including the possibility of fine and imprisonment.

Dana Nissen

SIGNATURE OF OWNER OF FACILITY

SIGNATURE OF OPERATOR OF FACILITY

Nissen Dairy, Inc.

PRINT OR TYPE NAME

SAME AS OWNER

PRINT OR TYPE NAME

6-18-24

DATE

DATE

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217 Primo Way • Modesto, California 95358 • Office (209) 538-8111 • FAX (209) 538-3966

Nissen Dairy

17283 S. Brennan
Escalon,CA 95320

Report # L2.2-14R64081
Report Date: 06/26/2023
Received Date: 06/14/2023
Work Order: 603457

Laboratory Number: 60345703
Description: Water, S.S.J.I.D. Irrigation, 06-14-23, 1400

Analytical Results

Constituent	Result		Reporting Limit	Method Detection Limit	Method Reference	Analysis Date	Analysis Time	Analyst
Electrical Conductivity (E.C.)	74	umho/cm	1	1	SM 2510B[5]	06/20/2023	13:00	VM
Total Dissolved Solids (TDS)	61	mg/l	20	7	SM 2540C	06/16/2023	11:20	JA
Nitrate Nitrogen	< 0.4	mg/l	0.4	0.01	EPA 300.0	06/15/2023	09:45	JDLC
Total Kjeldahl Nitrogen (TKN)	< 2	mg/l	2	0.8	SM 4500-Norg B	06/20/2023	08:30	JN
Total Nitrogen	< 2.4	mg/l			Calculation	06/22/2023	NA	JN

QC Results

Constituent	QC Units	Blank	Matrix	Theoretical Spike	Matrix Spike	Matrix Spike Duplicate	Lab Control Spike	Precision %	Accuracy %	Respons %
Electrical Conductivity (E.C.)	umho/cm	< 1	74	1400	1400	1400	1400	0.1	96.9	100.
Total Dissolved Solids (TDS)	mg/l	< 20	360	3000	3400	3400	3000	0.3	100.8	100.
Nitrate Nitrogen	mg/l	< 0.1	13	0.9	14	14	0.9	0.0	97.3	97.
Total Kjeldahl Nitrogen (TKN)	mg/l	< 2	< 2	28	29	28	28	2.5	99.9	98.
Total Nitrogen	mg/l	NA	NA	NA	NA	NA	NA	NA	NA	N

Notes

5 At 25 °C

Authorized By: _____ 06/26/2023

Amos Snider Laboratory Supervisor

ELAP Accreditation Laboratory Certificate #2776. Unless otherwise noted, all samples were received in acceptable condition. The result(s) in this rep
relate only to the portion of the sample(s) tested. Results are not corrected for recovery or expanded measurment of uncertainty. This report does n
constitute a release of product for consumption. This report shall not be reproduced except in full, without written approval of the laboratory. This

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Nissen Dairy

17283 S. Brennan
Escalon,CA 95320

Report # L2.2-14R64081
Report Date: 06/26/2023
Received Date: 06/14/2023
Work Order: 603457

Laboratory Number: 60345702
Description: Water, Irrigation Well, 06-14-23, 1400

Analytical Results

Constituent	Result		Reporting Limit	Method Detection Limit	Method Reference	Analysis Date	Analysis Time	Analyst
Electrical Conductivity (E.C.)	710	umho/cm	1	1	SM 2510B[5]	06/20/2023	13:00	VM
Total Dissolved Solids (TDS)	540	mg/l	20	7	SM 2540C	06/16/2023	11:20	JA
Nitrate Nitrogen	13	mg/l	0.1	0.01	EPA 300.0	06/15/2023	09:45	JDLC
Total Kjeldahl Nitrogen (TKN)	< 2	mg/l	2	0.8	SM 4500-Norg B	06/20/2023	08:30	JN
Total Nitrogen	13.0	mg/l			Calculation	06/22/2023	NA	JN

QC Results

Constituent	QC Units	Blank	Matrix	Theoretical Spike	Matrix Spike	Matrix Spike Duplicate	Lab Control Spike	Precision %	Accuracy %	Response %
Electrical Conductivity (E.C.)	umho/cm	< 1	74	1400	1400	1400	1400	0.1	96.9	100.(
Total Dissolved Solids (TDS)	mg/l	< 20	360	3000	3400	3400	3000	0.3	100.8	100.(
Nitrate Nitrogen	mg/l	< 0.1	13	0.9	14	14	0.9	0.0	97.3	97.!
Total Kjeldahl Nitrogen (TKN)	mg/l	< 2	< 2	28	29	28	28	2.5	99.9	98.!
Total Nitrogen	mg/l	NA	NA	NA	NA	NA	NA	NA	NA	N/

Notes

5 At 25 °C

Authorized By: _____ 06/26/2023
Amos Snider Laboratory Supervisor

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Nissen Dairy

17283 S. Brennan
Escalon,CA 95320

Report #	L2.2-14R64081
Report Date:	06/26/2023
Received Date:	06/14/2023
Work Order:	603457

Laboratory Number: 60345701
Description: Water, Well, 06-14-23, 1400

Analytical Results

Constituent	Result		Reporting Limit	Method Detection Limit	Method Reference	Analysis Date	Analysis Time	Analyst
Electrical Conductivity (E.C.)	710	umho/cm	1	1	SM 2510B[5]	06/20/2023	13:00	VM
Total Dissolved Solids (TDS)	500	mg/l	20	7	SM 2540C	06/16/2023	11:20	JA
Nitrate Nitrogen	13	mg/l	0.1	0.01	EPA 300.0	06/15/2023	09:45	JDLC
Total Kjeldahl Nitrogen (TKN)	< 2	mg/l	2	0.8	SM 4500-Norg B	06/20/2023	08:30	JN
Total Nitrogen	13.0	mg/l			Calculation	06/22/2023	NA	JN

QC Results

Constituent	QC Units	Blank	Matrix	Theoretical Spike	Matrix Spike	Matrix Spike Duplicate	Lab Control Spike	Precision %	Accuracy %	Response %
Electrical Conductivity (E.C.)	umho/cm	< 1	74	1400	1400	1400	1400	0.1	96.9	100.(
Total Dissolved Solids (TDS)	mg/l	< 20	360	3000	3400	3400	3000	0.3	100.8	100.(
Nitrate Nitrogen	mg/l	< 0.1	13	0.9	14	14	0.9	0.0	97.3	97.!
Total Kjeldahl Nitrogen (TKN)	mg/l	< 2	< 2	28	29	28	28	2.5	99.9	98.7
Total Nitrogen	mg/l	NA	NA	NA	NA	NA	NA	NA	NA	N/

Notes

5 At 25 °C

Authorized By: _____ 06/26/2023

Amos Snider Laboratory Supervisor

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Company Name: NISSEN DAIRY
Attn:
Address:
Phone:
Fax:
Email:

JL Analytical Services
217 Primo Way
Modesto, CA 95358
Phone (209) 538-8111
FAX (209) 538-3966

2F077 ver. 3 ELAP chain of custody

Analysis Requested (write in the tests)

Lab Use Only: Work Order Number	Matrix	Sample Description	Sampling Date	Sampling Time	Chlorine Level	Field pH	Sample Type	Reason For Analysis	Sample Preserv.	ELECTRICAL CONDUCTIVITY	TOTAL DISSOLVED SOLIDS	TOTAL NITROGEN						Temp as read	Corr. Temp
		Sample Information:								Check the Appropriate Box Below:									
60345701		WELL WATER	6-14-23	1:00			W	A		X	X	X						25.2	25.3
02		IRRIGATION WELL WATER	6-14-23	1:00			W	A		↓	↓	↓						25.3	25.4
03		S.S.I.D. IRRIGATION WATER	6-14-23	1:00			D	A		↓	↓	↓						24.9	25.0

Thermometer ID: IR2

Sampled by: DANA NISSEN	Received by (signature):		Date:	Time:
Relinquished by (signature): Dana Nissen	Received by (signature): Kelsey Wagle		6/14/23	1400
Relinquished by (signature):			Date:	Time:

Notes / Special Instructions:

DAIRY QUALITY

Key:

Sub sample bottle ID: If more than one bottle per sample assign a unique letter to each (A, B, C, etc.)

Matrix: DW=Drinking Water GW=Ground Water SW=Storm Water WW=Waste Water

Sample Type: D=Distribution W=Well O=Other

Reason for Analysis: A=Routine B=Repeat C=Special D=Other

Sample Preserv.: M: Cool, <10 deg C, Sodium Thiosulfate ($Na_2S_2O_3$), C: </= 6 deg C NA: Nitric Acid (HNO_3) SA: Sulfuric Acid (H_2SO_4) H: Hydrochloric Acid (HCl)